SIADS 696 Milestone II Project Report - Team 06

Project name: Do Tumor-infiltrating Immune Cell Proportions Improve Three-year Overall Survival Prediction Beyond Clinical Variables in Kidney Renal Clear Cell Carcinoma (KIRC)?

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1. Introduction:

Kidney Renal Clear Cell Carcinoma (KIRC) is a common type of kidney cancer. Clinical variables such as age, pathologic stages, and histological grade are commonly used to estimate patient prognosis. However, patients with similar clinical characteristics may still have different survival outcomes. This suggests that biological information from the tumor microenvironment may provide additional information about patient risk.

Tumor-infiltrating immune cells are an important part of the tumor microenvironment. Some immune cells may support an antitumor response, while others may contribute to immune suppression and tumor progression. Previous studies have used CIBERSORT to estimate immune cell proportions from tumor gene-expression data and have reported relationships between several immune cell types and survival in renal cell carcinoma (Zhang et al., 2019; Pan et al., 2020). However, an immune feature associated with survival may not necessarily improve patient-level prediction after established clinical variables are considered.

The main question of this project is whether tumor-infiltrating immune cell proportions improve the prediction of three-year overall survival in KIRC beyond established clinical variables. Answering this question may help evaluate whether transcriptome-based immune estimates provide useful information for patient risk assessment. Even if immune features do not improve prediction, this result can still help explain the practical value and limitations of transcriptome-based immune estimates.

We combined The Cancer Genome Atlas (TCGA) clinical and survival information with CIBERSORT estimates of 22 immune cell populations. The supervised learning analysis compared four feature representations: clinical only, immune only, clinical plus immune, and clinical plus PCA-transformed immune features. Logistic Regression, Support Vector Machine, and Random Forest represented three different model families. Five-fold nested stratified cross-validation was used for evaluation, with three-fold inner cross-validation for hyperparameter tuning. The same patients and outer folds were used for all feature representations. All imputation, scaling, categorical encoding, and PCA steps were fitted only within the training folds. In the unsupervised analysis, immune cell proportions were used to identify major immune patterns and possible patient subgroups. Clinical variables and survival outcomes were not used to create or select the clusters, but were used later to describe the identified groups.

Compared with earlier studies, we focused on the direct evaluation of whether CIBERSORT immune cell proportions provide predictive information beyond routine clinical variables. The clinical-only and clinical-plus-immune models used the same patients and cross-validation folds, allowing a direct comparison between the two feature sets. The unsupervised analysis provided a second view of the data by examining whether patients could be separated into stable immune subgroups.

The supervised analysis showed that the clinical-only Random Forest achieved the highest mean ROC-AUC of 0.805 ± 0.048. Adding the original immune features reduced the mean ROC-AUC for all three model families. PCA improved some models compared with the original immune representation, but it did not outperform the clinical-only models. Therefore, the CIBERSORT immune-cell proportions did not provide consistent additional predictive value beyond the selected clinical variables.

The unsupervised analysis identified two overlapping immune groups rather than tight well separated subgroups (k means silhouette score 0.127 and hierarchical clustering 0.113). One group had higher CD8 T-cell proportions and more stage IV and grade 4 tumors, while the other had higher M2 macrophage proportions and more stage I tumors. These clusters differed in stage and grade, but they did not show different overall survival (log-rank $p = 0.34$ for k-means and $p = 0.31$ for hierarchical clustering).

2. Related Work:

Zhang et al. (2019) used TCGA gene expression data and CIBERSORT to estimate the proportions of 22 tumor-infiltrating immune cell types in 891 renal cell carcinoma tumors. They examined the relationships between individual immune cell proportions and overall survival using Cox regression. Our project uses similar TCGA and CIBERSORT data, but we focus on patient-level prediction rather than mainly testing individual immune-survival associations. We directly compare clinical-only and clinical-plus-immune models to determine whether immune features provide additional value for predicting three-year survival.
Pan et al. (2020) used TCGA and GEO data to compare immune cell infiltration between KIRC and normal kidney tissues. They examined the associations among immune cells, clinical characteristics, and survival outcomes, but none of the immune cell types remained an independent prognostic factor after established clinical variables were considered. Our project directly compares the predictive performance of four feature sets using the same patients and outer cross-validation folds instead of mainly testing statistical associations.
Zheng et al. (2021) used CIBERSORT and consensus clustering to identify four immune infiltration clusters in KIRC and examined their relationships with patient prognosis. Their study provided a broad description of the immune microenvironment. Our study has a narrower prediction question: whether CIBERSORT immune proportions provide additional predictive value beyond age, sex, stage, and grade for three-year survival.

3. Data Sources:

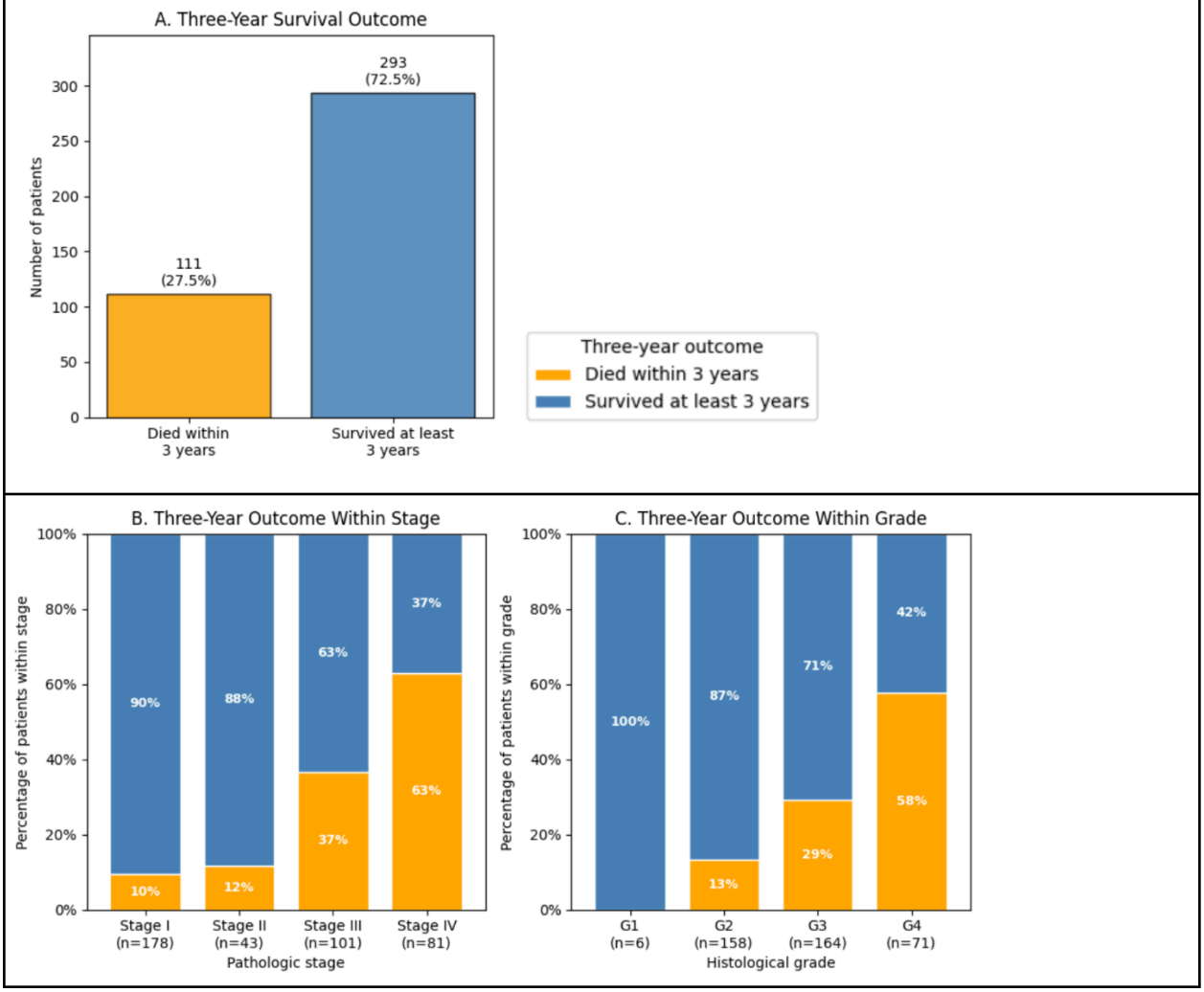
This project used two publicly available datasets from The Cancer Genome Atlas (TCGA). The TCGA Clinical Data Resource was downloaded from the NCI Genomic Data Commons Pan-Cancer Clinical Data Resource page: https://gdc.cancer.gov/about-data/publications/PanCan-Clinical-2018. It was provided as an Excel workbook (TCGA-CDR-SupplementalTableS1.xlsx). It contains standardized clinical and survival information for more than 11,000 patients across 33 TCGA cancer types (Liu et al., 2018). After selecting records with a cancer type of KIRC, 537 clinical patients remained. Important clinical variables included age, sex, pathologic stage, histological grade, overall survival status, and survival time. Survival time was measured in days.The immune data were downloaded from the NCI Genomic Data Commons PanImmune page: https://gdc.cancer.gov/about-data/publications/panimmune. It was provided as a tab-separated file (TCGA.Kallisto.fullIDs.cibersort.relative.tsv). The PanImmune project analyzed immune characteristics across 33 TCGA cancer types (Thorsson et al., 2018) and contained 11,373 samples and 27 variables. The CIBERSORT file used in this project contained relative proportions for 22 immune cell populations and three quality measures: P-value, correlation, and root mean squared error (RMSE). After selecting KIRC records, 618 immune samples remained.
The two datasets were matched by TCGA patient ID. After the patient- and sample-level processing described in the next section, 533 patients had matched clinical and immune information. Among these patients, 404 had a known three-year survival outcome and were used for supervised learning. The supervised cohort included 111 patients who died within three years and 293 patients who survived for at least three years.

4. Feature Engineering:

The clinical and immune datasets were first limited to KIRC records. The TCGA sample barcode was standardized by replacing periods with hyphens. The first three barcode sections were used to create the patient ID, and the sample-type code was extracted from the barcode. Only samples with code 01, which represents primary tumor tissue, were retained. This step produced 545 primary tumor samples from 533 unique patients.
For patients who had more than one primary tumor sample, only one sample was selected without using clinical outcomes. Samples were ranked by lower CIBERSORT p-value, lower RMSE, and higher correlation. Sample ID was used as a final tie-breaker. Survival information was not used during this selection process.
The clinical and immune datasets were then joined by patient ID using an inner merge. The merge was validated as a one-to-one match. The resulting dataset contained 533 unique patients, and no patient appeared more than once.
The clinical variables were examined for actual missing values and text values that represented unavailable information. The values “Not Available”, “Discrepancy”, and “GX” were converted to missing values. Age, overall survival status, and overall survival time were converted to numeric variables, with

invalid entries converted to missing values. In the final supervised cohort, age , sex and 22 immune cell proportions had no missing values. The pathologic stage had 1 missing value, and the histological grade had 5 missing values. Missing values were not filled before cross-validation. Within each training fold, numerical variables were imputed using the median, while categorical variables were imputed using the most frequent value.
A three-year cutoff was defined as 1,095 days. Patients with OS = 1 and OS.time less than or equal to 1,095 days were labeled as dying within three years (survival_3y = 0). Patients whose observed survival or follow-up time equaled or extended beyond 1,095 days were labeled as surviving for at least three years (survival_3y = 1). Patients who did not meet either rule were excluded from supervised learning because their three-year status could not be determined without making assumptions about their later outcome. This process resulted in 404 patients with a valid binary outcome. The OS and OS.time variables were used only to construct the outcome and were not included as prediction features. Patient ID, sample ID, cancer type, sample type, overall survival status, overall survival time, and the CIBERSORT quality measures were not used as prediction features. This prevented the models from using patient identifiers, sample metadata, or variables that directly defined the outcome.

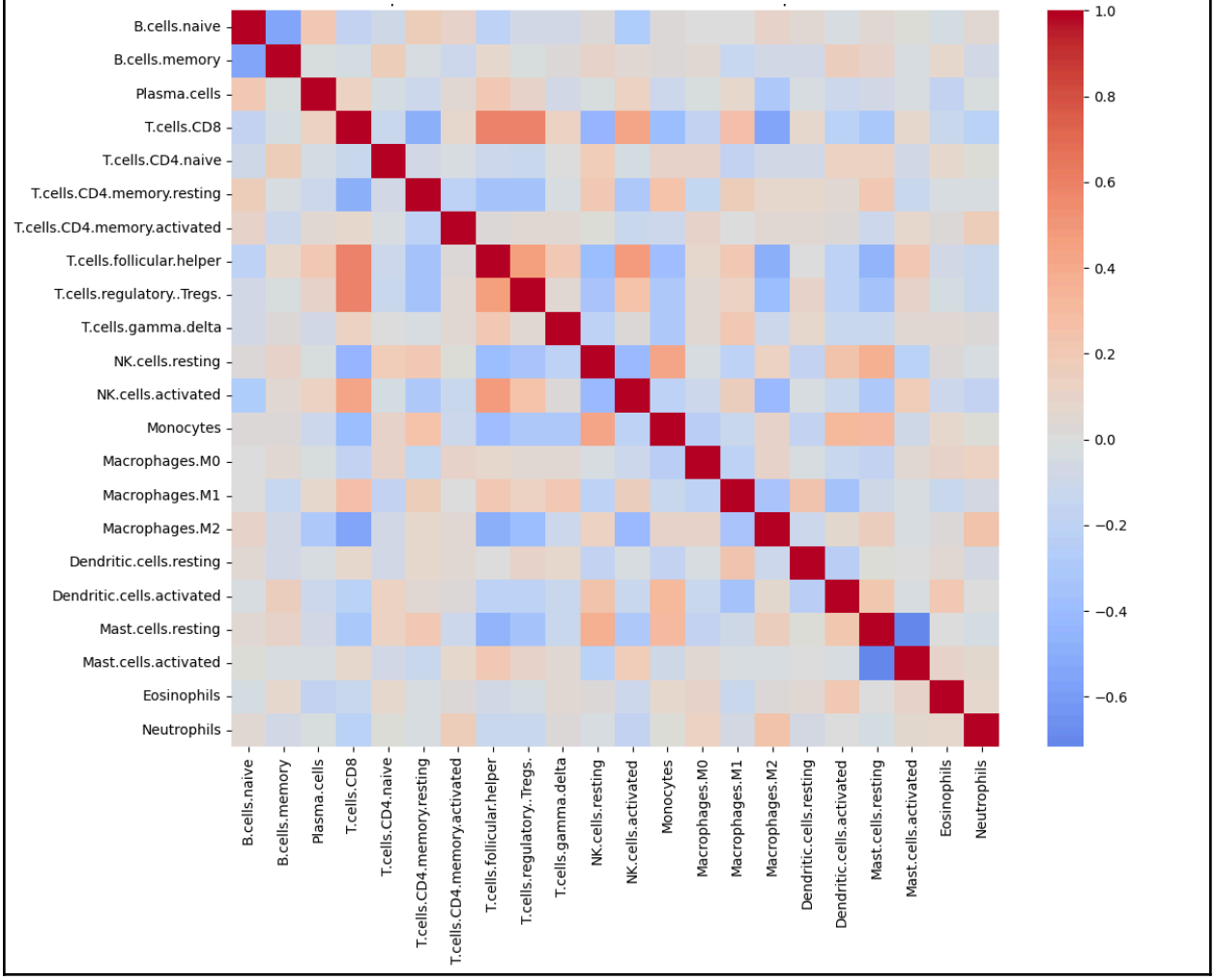
Figure 1: Distribution of clinical characteristics and three-year survival outcomes



Four feature representations were evaluated. The clinical-only representation contained four original variables: age, sex, pathologic stage, and histological grade. The immune-only representation contained the 22 CIBERSORT immune-cell proportions. The clinical-plus-immune representation contained all 26

original variables. The PCA representation retained the clinical variables and replaced the 22 immune variables with principal components explaining 90% of the immune-feature variance.
All preprocessing was included inside the modeling pipelines. Categorical variables were one-hot encoded, and numerical variables were standardized. When PCA was used, scaling and PCA were fitted only on the training data. Therefore, information from the outer test folds was not used during preprocessing or model training.
Figure 1
shows that the survival outcomes were imbalanced. The proportion of patients who died within three years generally increased at higher pathologic stages and histologic grades.
Figure 2 shows that several immune cell proportions were moderately or strongly correlated. This correlation suggests that some immune variables may contain overlapping information. The clinical-plus-immune set contained all 32 predictors.

Figure 2: Spearman correlation among the 22 CIBERSORT immune cell proportions



5. Part A: Supervised Learning:

5.1 Methods:

We evaluated three different model families. Logistic Regression was used as a simple and interpretable linear baseline. SVM was selected as a margin-based method that can model linear or nonlinear decision boundaries. Random Forest was selected as a tree-based method that can capture nonlinear relationships and feature interactions.
We used five-fold nested stratified cross-validation. The outer folds were used for evaluation, while three-fold inner cross-validation and GridSearchCV were used for hyperparameter selection. Logistic

Regression tested C values of 0.1, 1, and 10. SVM tested the same C values with linear and radial basis function kernels. Random Forest used 300 trees and tested maximum depths of None, 5, and 10 and minimum leaf sizes of 1 and 5. Both no class weighting and balanced class weighting were considered. Hyperparameters were selected using ROC-AUC.

5.2 Evaluation Metrics:

ROC-AUC was selected as the primary metric because it evaluates how well a model ranks survivors and deaths across different thresholds. Because only 27.5% of the patients died within three years, balanced accuracy and death recall were also important. Survival average precision, precision, recall, and F1 score were reported as secondary metrics. Results were summarized using the mean and standard deviation across the five outer folds.

5.3 Overall results:

The clinical-only Random Forest achieved the highest mean ROC-AUC, while the clinical-only SVM achieved the highest balanced accuracy. For all three model families, the clinical-only representation had the highest mean ROC-AUC. Adding the original immune features changed ROC-AUC by −0.018 for Logistic Regression, −0.014 for Random Forest, and −0.050 for SVM. Improvement occurred in only two of five folds for Logistic Regression and one of five folds for Random Forest and SVM. (**Table 1**)

***Table 1**: Best feature representation from each model family*

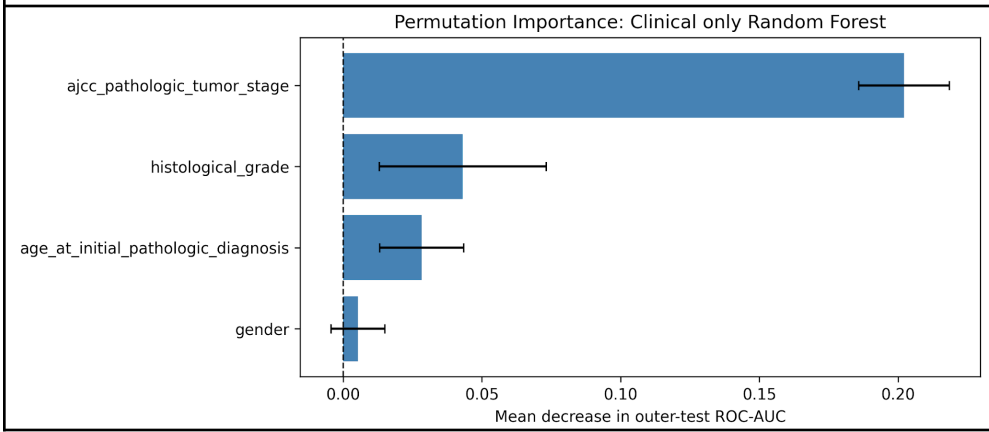
Model	Best feature set	ROC-AUC	Survival AP	Balanced accuracy
Logistic Regression	Clinical only	0.787 ± 0.044	0.896 ± 0.028	0.707 ± 0.079
SVM	Clinical only	0.776 ± 0.033	0.879 ± 0.025	0.734 ± 0.038
Random Forest	Clinical only	0.805 ± 0.048	0.909 ± 0.026	0.719 ± 0.054

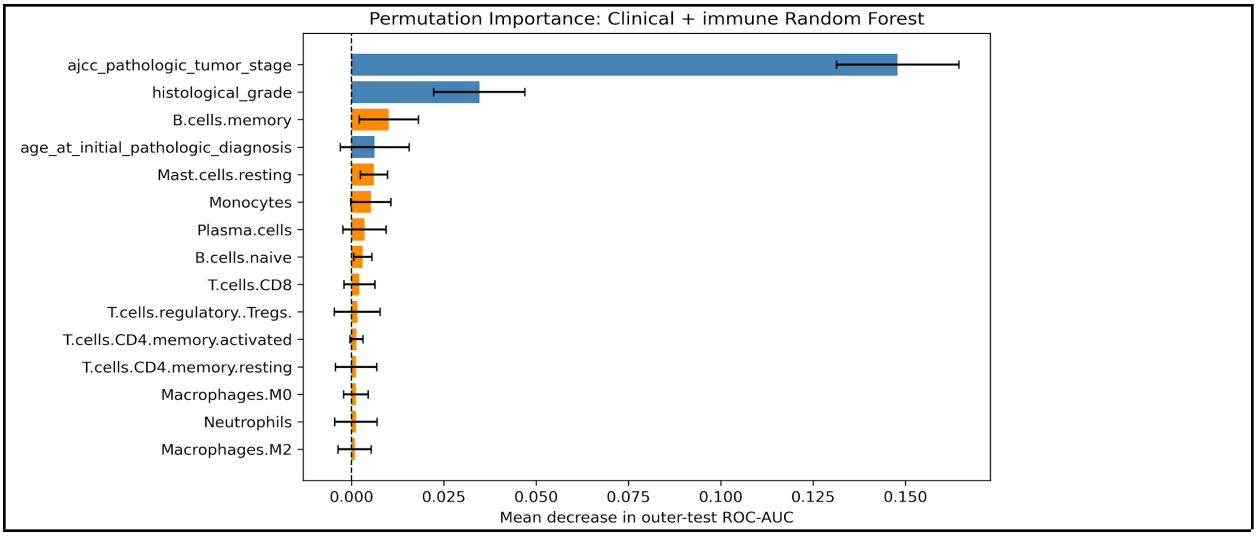
5.4 Feature Importance:

Permutation importance was calculated for the clinical-only and clinical-plus-immune Random Forest models using the outer test folds. Each original feature was shuffled 30 times, and the resulting decrease in ROC-AUC was measured. A larger decrease indicated that the model depended more strongly on that feature.

The pathologic stage was the most important feature. Its mean permutation importance was 0.203 in the clinical-only model and 0.144 in the clinical-plus-immune model. Histological grade and age had importance values of 0.044 and 0.030 in the clinical-only model. Among the immune variables, memory B cells had the highest mean importance at 0.011, followed by resting mast cells at 0.007. Most other immune features had values close to zero or below zero. (**Figure 3**)

***Figure 3**: Random Forest permutation importance for the clinical-only and clinical-plus-immune model.*





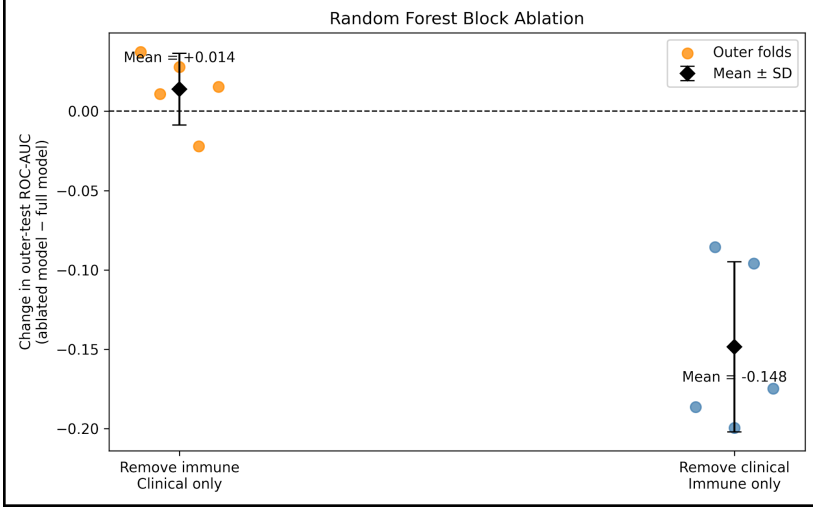
5.5 Ablation Analysis:

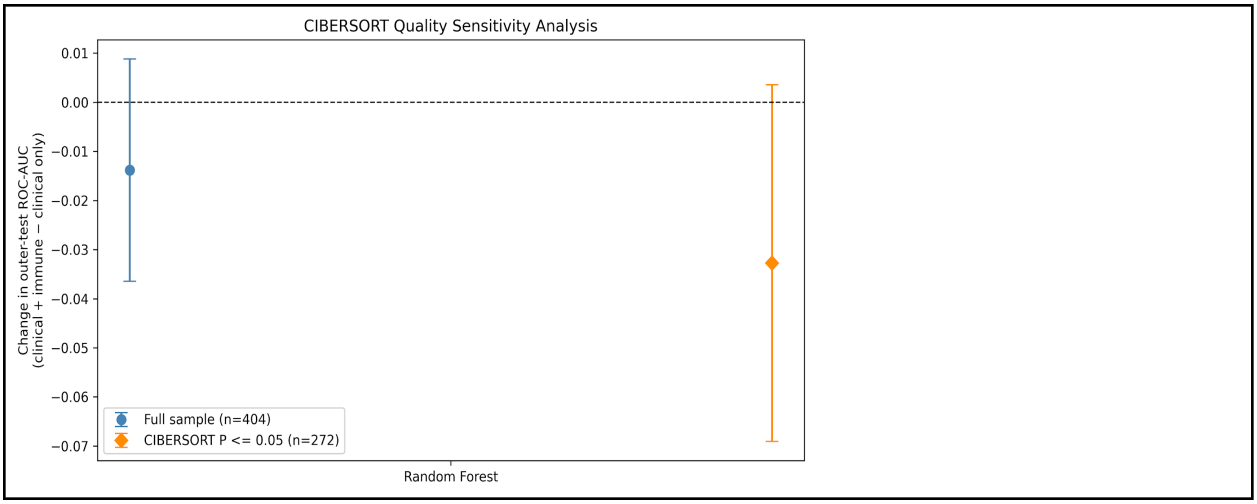
Block ablation compared the full clinical-plus-immune Random Forest with models that removed either the immune or clinical feature block. The full model had a mean ROC-AUC of 0.791, survival AP of 0.902, and balanced accuracy of 0.641. After removing the immune features, these values increased to 0.805, 0.909, and 0.719. After removing the clinical features, they decreased to 0.643, 0.822, and 0.505. Removing the immune block increased ROC-AUC by an average of 0.014 and improved four of five outer folds. Removing the clinical block reduced ROC-AUC by 0.148 and reduced performance in all five folds. This suggests that the clinical variables provided most of the predictive information. (**Figure 4, upper**)

5.6 Sensitivity Analysis:

We repeated the Random Forest analysis after limiting the cohort to patients with a CIBERSORT P-value of 0.05 or lower. This reduced the sample size from 404 to 272 patients. In this subset, the clinical-only model achieved a mean ROC-AUC of 0.832 ± 0.069, compared with 0.799 ± 0.051 for the clinical-plus-immune model and 0.809 ± 0.071 for the PCA model. Adding the original immune features changed ROC-AUC by −0.033 ± 0.036 and improved only one of five folds. Therefore, the stricter quality filter did not reverse the main conclusion. (**Figure 4, lower**)

***Figure 4**: Random Forest block ablation and CIBERSORT quality sensitivity analysis*





5.7 Tradeoffs:

The classification threshold created a tradeoff between identifying survivors and identifying deaths. At a threshold of 0.30, survivor recall was 0.925, but death recall was only 0.378. At a threshold of 0.60, survivor recall decreased to 0.734, while death recall increased to 0.748 and balanced accuracy reached 0.741. Therefore, a higher threshold improved the detection of deaths but missed more survivors. The threshold analysis was exploratory and was not used to replace the primary nested cross-validation results. Random Forest had the highest ROC-AUC, while SVM had the highest balanced accuracy. (**Figure 5**)

5.8 Failure Analysis of the Random Forest Model:

The clinical-only Random Forest produced 98 incorrect outer-fold predictions. Case A had favorable Stage I and G2 characteristics, which may have caused the model to overestimate survival. Replacing G2 with the training-fold reference grade reduced the survival score by 0.165. This patient shows that an apparently favorable clinical profile does not guarantee long survival.

Case B had Stage IV and G4 disease, but survived for at least 2,227 days. Replacing Stage IV with Stage I increased the predicted survival score by 0.517, showing that stage strongly influenced this incorrect prediction. This case represents a patient who survived longer than expected from the clinical variables.

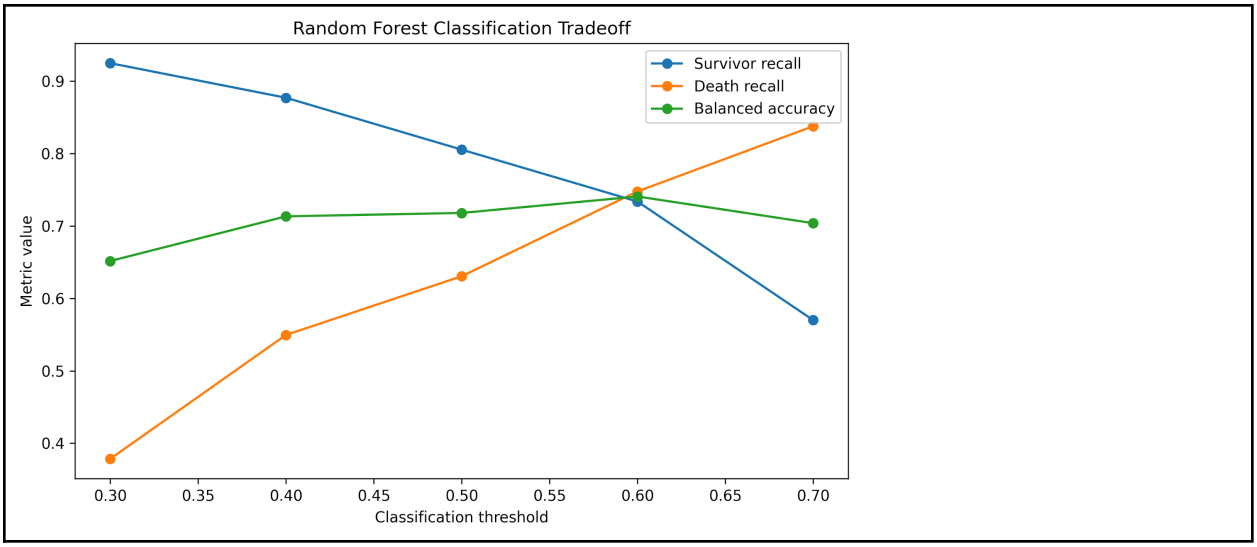
Case C had a missing pathologic stage. The pipeline imputed this value using the most frequent stage in the training fold, but the patient's actual disease extent was unknown. This missing information may have limited the model's ability to estimate risk correctly. (**Table 2**)

Future work could include treatment information, comorbidities, molecular features, and more detailed tumor characteristics. A missing-value indicator may also help the model distinguish missing stage information from an observed low stage. Time-to-event survival models and external validation could further improve the analysis.

***Table 2**: Three specific Random Forest prediction failures*

Case	Patient	Actual outcome	Predicted outcome	Survival score	Clinical profile
A	TCGA-BP-4353	Died within 3 years	Survived	0.942	Age 61, male, Stage I, G2; died at 375 days
B	TCGA-CJ-4637	Survived at least 3 years	Died	0.084	Age 52, female, Stage IV, G4; follow-up 2,227 days
C	TCGA-BP-4798	Died within 3 years	Survived	0.561	Age 74, male, stage missing, G4; died at 334 days

***Figure 5**: Tradeoff between survivor recall, death recall, and balanced accuracy across classification thresholds.*



5.9 Failure Analysis of the Logistic Regression Model

To better understand the limitations of our machine learning model, we conducted a failure analysis by examining incorrectly classified samples and identifying the features that contributed most strongly to these predictions. False positive and false negative cases were analyzed separately, and feature contributions were compared between correctly and incorrectly classified samples to identify potential patterns associated with prediction errors.

The Logistic Regression model was evaluated using a held-out test dataset consisting of 80 patient samples. A total of 14 samples were incorrectly classified, including 4 patients who were incorrectly predicted to die within three years (false negatives) and 11 patients who were incorrectly predicted to survive (false positives). The classification results are summarized in **Table 3 (**Attached File: Table 1 Logistic_Regression_Failure_Analysis.xlsx**)**.

5.9.1 Priority Analysis

To estimate the potential impact of future improvements on model performance, the held-out test dataset was divided into four patient subgroups based on clinical characteristics and feature distributions (**Table 4**). Group 1 included patients aged 70 years or older, while Group 2 consisted of patients diagnosed with AJCC Stage IV cancer. Group 3 included patients with unusual immunological measurements, defined as having at least one immune-cell feature outside the 10th–90th percentile range of the training data. Group 4 consisted of patients with extreme feature values, defined as having at least three numerical features outside the 1st–99th percentile range of the training data.
For each subgroup, the number of incorrectly classified patients was calculated, along with the percentage of the entire test dataset represented by these errors, using the following formula:

$$PercentofTestData = \frac{IncorrectPatientsinGroup}{AllTestPatients} \times 100$$

The subgroup failure analysis revealed that patients with unusual immune-cell measurements accounted for the largest proportion of incorrect predictions, representing **13.58% of the entire test dataset**. Patients aged 70 years or older accounted for an additional 8.64% of the test dataset through incorrect predictions. Interestingly, no patients met the predefined criteria for extreme feature values, suggesting that this category did not contribute to the observed prediction errors under the current definition.

Table 4 *The grouped features impact on model failure*

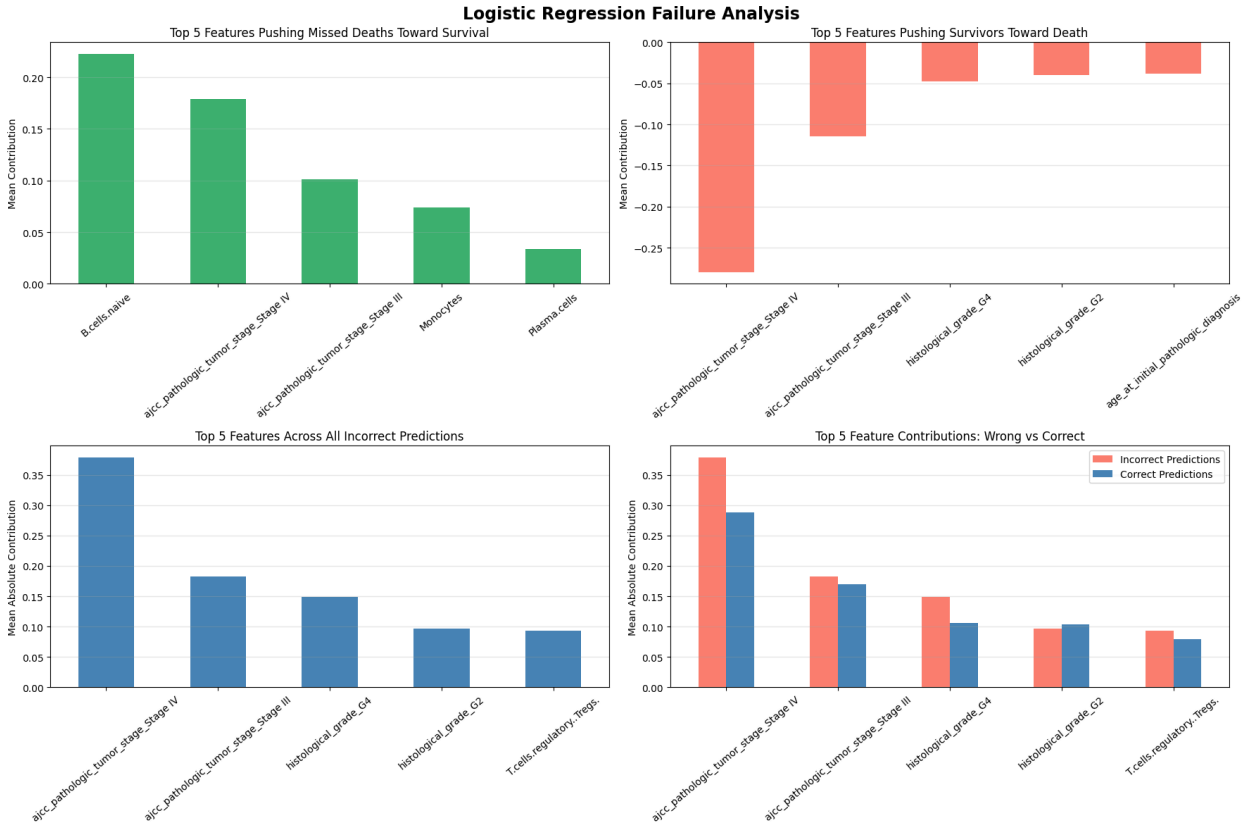
index	Grouped Features	Affected Patients	% of Test Data	Priority
1	Group with unusual immune-cell values	11	13.58	High

2	Group with age > 70 +	7	8.64	Medium
0	Group with Stage IV patients	5	6.17	Medium
3	Group with unusual records	0	0.0	Low

5.9.2 Feature selection and model performance

Next step we want to understand which specific feature will contribute more to the model failure in prediction. Logistic Regression coefficients indicate the direction and strength of each feature's association with the predicted outcome. For individual patients, feature contributions can be calculated by multiplying the standardized feature values by their corresponding model coefficients.
Although feature contributions do not directly establish the underlying causes of prediction errors, analyzing these contributions in misclassified samples can provide valuable insights into potential sources of model failure. By identifying features that strongly influence incorrect predictions, this analysis can help reveal patterns associated with misclassification, guide further investigation into possible root causes, and identify opportunities for improving feature representation and model performance.

Figure 6: Top five features push your Logistic Regression model toward predicting death in failure cases .



6. Part B: Unsupervised Learning:

6.1 Overview

The unsupervised analysis used all 533 patients with a matched KIRC primary tumor immune profile,including patients whose three year outcome was unknown. Age, sex, stage, grade, and survival were not used to create the features or to choose the number of groups. Three of the 22 CIBERSORT cell types were zero in at least 90% of patients and were removed before clustering: naive CD4 T cells, activated memory CD4 T cells, and eosinophils. The remaining 19 proportions were rescaled so that each

patient summed to 1. Remaining zeros were replaced with a small positive value, a centered logratio transformation was applied, and the features were standardized. Kmeans and hierarchical clustering with Ward linkage were fit to these 19 features using Euclidean distance. Each method was run for 2 through 8 groups. The selected solution was the one with the highest silhouette score in which every group contained at least 5% of the patients, or 27 patients. Silhouette, CalinskiHarabasz, and DaviesBouldin scores were recorded with the group sizes. Stability of the selected kmeans solution was assessed with 50 samples of 80% of the patients.In each sample the transformation and kmeans fit were repeated, and agreement with the full data labels was measured by the Adjusted Rand Index. After the groups were fixed, their immune composition, stage,grade, and Kaplan Meier survival curves were compared. A logrank test was used for the survival comparison. As specified in the proposal, principal component analysis was also applied to the same 19 standardized features. The smallest number of components explaining at least 80% of the variance was retained and both clustering methods were repeated on those scores at the end of the code.

6.2 Results

Both methods selected two groups, but the separation was weak. Kmeans assigned 301 patients to one group and 232 to the other. The silhouette score was 0.127, the CalinskiHarabasz score was 75.8, and the DaviesBouldin score was 2.56. Solutions with four or more groups left at least one group below the 5% size rule. Hierarchical clustering assigned 313 and 220 patients to the two groups and had a silhouette score of 0.113. Its threegroup solution had a silhouette score of 0.121, but one group contained only 14 patients and was rejected. The Adjusted Rand Index between kmeans and hierarchical clustering was 0.64, and 54 patients were assigned to different groups. The 50 repeated samples agreed with the full kmeans labels at a mean Adjusted Rand Index of 0.93 (minimum 0.75). The two group division was reproducible but the patients overlapped rather than forming two tight groups.

The two methods described the same immune pattern. One group was higher in CD8 T cells, and the other was higher in M2 macrophages. In the kmeans solution, the larger group was 6.0 percentage points above the cohort mean for CD8 T cells and 3.5 points below the mean for M2 macrophages. The smaller group was 7.8 points below the mean for CD8 T cells and 4.5 points above the mean for M2 macrophages. The CD8 higher group contained more stage IV tumors (60 of 301, 19.9%) and more grade 4 tumors (57 of 301,18.9%). The macrophagehigher group contained more stage I tumors (138 of 232, 59.5%) and fewer grade 4 tumors (19 of 232, 8.2%). Stage and grade differed between the kmeans groups (chisquare $p = 0.0004$ for stage and $p < 0.001$ for grade). Overall survival did not differ. The kmeans logrank pvalue was 0.34, and the median survival times were 7.7 and 7.5 years. The hierarchical logrank pvalue was 0.31. One hierarchical group had a median survival of 7.6 years, and the median was not reached in the other group.

The PCA check kept 10 components, which explained 80.4% of the variance. Kmeans on the PCA scores again selected two groups, with 298 and 235 patients and a silhouette score of 0.151. Five patients changed groups relative to kmeans on the original 19 features (Adjusted Rand Index 0.96). Hierarchical clustering on the PCA scores also selected two groups, with 305 and 228 patients, and 42 patients changed groups (Adjusted Rand Index 0.71). The proposal's PCA step did not change the main finding. The immune profiles form a weak gradient that tracks tumor stage and grade. They do not identify survival defined subgroups.

***Table 5**: Group sizes, silhouette scores, and agreement*

	K-means, 19 features	Hierarchical, 19 features	K-means after PCA	Hierarchical after PCA

Group sizes	301 and 232	313 and 220	298 and 235	305 and 228
Silhouette	0.127	0.113	0.151	0.135
Patients who switch	—	54 (ARI 0.64)	5 (ARI 0.96)	42 (ARI 0.71)
Log-rank p-value	0.34	0.31	—	—

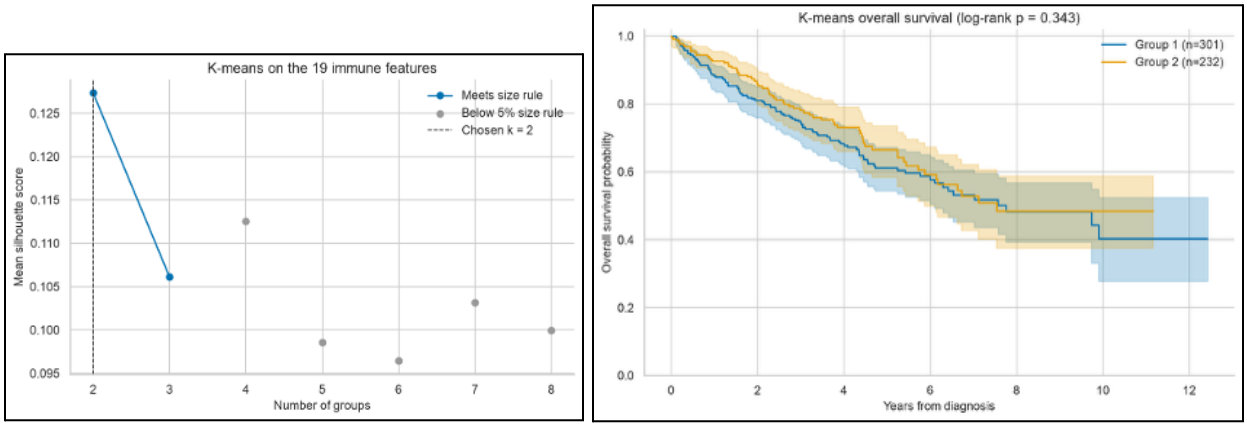


Figure 7: K-means silhouette scores on the 19 immune features (left), and Kaplan-Meier overall survival for the k-means groups (right)

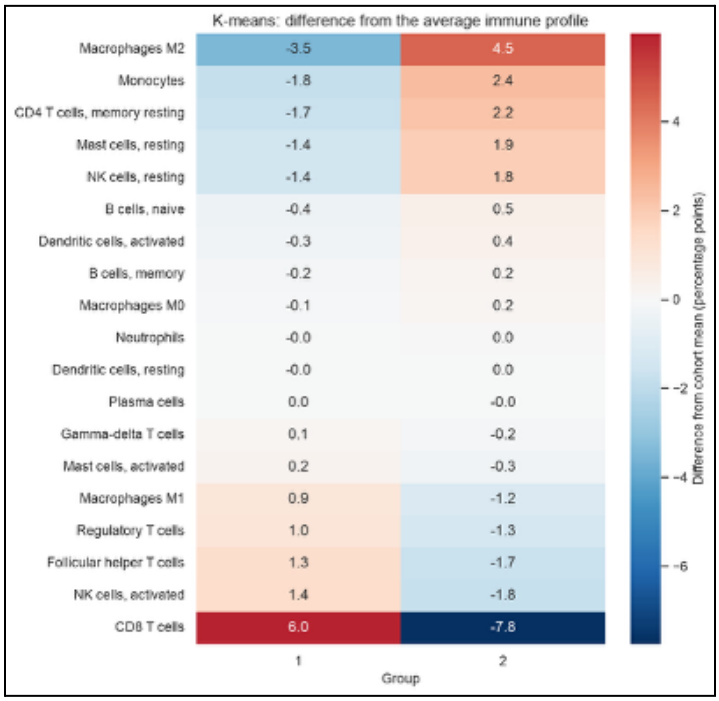


Figure 7: K-means difference from the average immune profile

6.3 Failure Analysis of Unsupervised Learning

The split into two groups is not a clean cut. 54 patients sit between the CD8 end and the macrophage end and change sides when the method changes. This is not ideal because the group averages then depend on where the cut is placed. A patient in the middle can be labeled either way. Keep the result as a spectrum and report the middle patients on their own.

The immune numbers are estimates and not cell counts. Zeros were filled in and three rare cell types were dropped. Patients alive for under three years were kept too. Most had stage I tumors and more sat on the macrophage end. This is not ideal because their unknown later survival can shape the stage gap and the survival curves. A p value of 0.34 does not prove the two ends live equally long. Repeat the checks on the 404 patients with a known three year outcome and test another way of filling zeros.

The same split returned in repeated samples and after PCA so it is stable in this dataset. It is still not ideal because all patients come from one public study. Hospitals and the amount of tumor in a sample can shift the immune numbers and can also follow stage. The link with stage may look stronger than it is in a new set of patients. We should repeat the spectrum at another hospital and record the hospital and the tumor amount.

7. Discussion:

Part A: Supervised Learning

The supervised analysis did not show that CIBERSORT immune-cell proportions improved three-year survival prediction beyond the selected clinical variables. The clinical-only Random Forest achieved the highest mean ROC-AUC of 0.805, while adding immune features reduced performance for all three model families. We were surprised that biologically relevant immune features provided little additional predictive value. The feature-importance and ablation results showed that the pathologic stage provided much more predictive information than the individual immune variables.

An important challenge was the relatively small and imbalanced dataset. The immune proportions were also correlated and may have added noise, which increased overfitting. We addressed these problems by using nested stratified cross-validation, class-weight tuning, PCA, and a stricter CIBERSORT quality filter. However, none of these analyses changed the main conclusion.

With more time and resources, we would validate the models in an independent patient cohort and include treatment, comorbidity, genomic, and molecular information. Time-to-event survival models could also use censored follow-up more efficiently than a binary three-year outcome. These extensions may help determine whether immune information is more useful in a larger and more complete clinical dataset.

Part B: Unsupervised Learning

The unsupervised analysis found an immune spectrum with two ends. One end has more CD8 cells and more advanced tumors. The other end has more M2 macrophages and more stage I tumors. The two ends overlap so these are not two clean groups. Where a patient sits on this spectrum lines up with how advanced the tumor is. Patients at both ends lived about the same length of time. The spectrum can describe the immune mix and its link with stage and grade. It does not tell how long an individual patient will live. That result fits the supervised models. The immune cell estimates did not add a clear survival signal beyond the clinical variables.

The hard part was that the split was weak. Patients overlapped and 54 patients changed groups when the method changed. The immune numbers are estimates and several cell types were mostly zeros. We kept age and sex and stage and survival out of the grouping so the groups were not built to match those outcomes. Repeating the analysis on smaller samples gave the same split. Compressing the features and clustering again also gave the same split. With more time we would repeat this in patients from another hospital. A larger set of patients could show whether this immune pattern holds outside this study.

Overall, the unsupervised analysis looked for immune groups without using age or sex or stage or grade or survival. Both methods found the same pattern. One end has more CD8 cells and more advanced

tumors. The other end has more M2 macrophages and more stage I tumors. The ends overlap and 54 patients change sides when the method changes. Patients at both ends lived about the same length of time.
The lesson is that the immune mix tracks how advanced the tumor is. It does not mark who lives longer. The useful result is a spectrum rather than two separate groups. A place on that spectrum describes the immune mix for a patient. It does not give a severity score or a survival prediction. This agrees with the supervised models because the immune estimates did not add a clear survival signal beyond the clinical variables. The same pattern returned in repeated samples and after PCA so it is real inside this dataset. It still needs a check in patients from another hospital before it is used more widely.

8. Ethical Consideration:

Part A: Supervised Learning

Incorrect survival predictions could affect clinical decisions if the model were used without careful review. A false prediction of poor survival could lead to unnecessary concern or aggressive treatment, while a false prediction of good survival could delay needed care. Therefore, this model should not be used as an independent clinical decision tool. Its predictions would need external validation and interpretation by healthcare professionals.
TCGA is a retrospective dataset and may not represent patients from other hospitals or demographic groups. Model performance should be evaluated across age, sex, racial, and clinical subgroups before clinical use. The CIBERSORT immune proportions are also estimated from gene-expression data rather than directly measured, which creates additional uncertainty.
Although the TCGA data are de-identified, patient privacy should still be protected when clinical and molecular datasets are combined. Future work should use secure data storage, limit access to patient-level records, and report only the information needed for analysis. Model limitations, uncertainty, and possible population bias should be clearly communicated to users.

Part B: Unsupervised Learning

The two immune groups should not be used to label a patient or to judge how severe the cancer is. The groups overlap and are only weakly separate. 54 patients fall into a different group when the second method is used. Many stage I tumors are in the group with more CD8 cells. Some stage IV tumors are in the group with more M2 macrophages. Patients in the two groups lived about the same length of time. Treating someone as higher risk because of their group could cause worry or change their care. The group does not tell us how long that person will live.

How the groups were built also matters. The immune cell numbers are estimates from gene data and not direct cell counts. Missing values were filled in and three rare cell types were left out. Different choices could move patients from one group to the other. These patients come from one public study and may not match patients at other hospitals. The pattern should be checked in new patients before it is used outside this study. Age and sex and stage and survival were not used to make the groups. They were used only later to describe them. A link with stage or grade does not mean the immune mix caused the tumor to be more advanced. The results include a file that connects each patient ID to a group. That file should be kept private. Reports should describe the groups as a whole and not one patient.

9. Statement of Work:

Qunshan Jia: Data curation on supervised data set. Train machine learning models including Logistic regression, Random forest, Support Vector Machine and conduct failure analysis on logistic regression as well as hierarchy bottom up clustering on supervised dataset

Fang Yuan: Cleaned and merged the clinical and immune datasets, performed exploratory data analysis, developed the supervised learning analysis, created the related tables and figures and wrote the introduction, related work, data sources, feature engineering, and supervised learning sections of the report.

Affan Atif:
Performed exploratory data analysis on the tumor datasets. Developed the unsupervised learning modelling, created related tables and figures, and compiled its discussion, analysis, overview, and strategy. Wrote sections regarding unsupervised learning of the report. Created the group GitHub repo to host the code files, figures, datasets, and README.

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